

THERMODYNAMIC CHARACTERISTICS OF THE INTERACTION OF FLUORAPATITE WITH LACTIC ACID

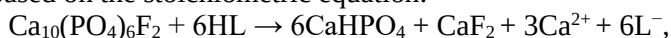
Provorkina A.A., Golovanova O.A.

Omsk State University

644077, Omsk, Mira ave., 55A

Poly(lactide) is one of the most widely used bioresorbable polymers in orthopedics and dentistry, serving as a matrix for implants and drug delivery systems. During its biodegradation, lactic acid is locally released, leading to a decrease in pH in the near-surface layer and initiating the dissolution of the mineral component of bone tissue [1]. Fluorapatite ($\text{Ca}_{10}(\text{PO}_4)_6\text{F}_2$, FAP) is a key inorganic constituent of tooth enamel and a promising material for implant coatings due to its high chemical stability. However, the thermodynamic aspects of its interaction with lactic acid under physiological conditions remain insufficiently understood [2].

Based on experimental data on the partial dissolution of fluorapatite in lactic acid solutions as a function of pH, and on the complete inhibition of dissolution at pH = 3.8, a thermodynamic model for the interaction between these compounds was developed. The model is based on the stoichiometric equation:



where HL denotes lactic acid and L^- denotes the lactate ion.

The developed thermodynamic model was experimentally validated and enables prediction of FAP behavior in acidic and neutral environments arising during poly(lactide) biodegradation and carbohydrate metabolism. It was established that the key factors limiting FAP dissolution are the formation of a passivating CaF_2 layer and the protonation of fluoride ions with the formation of HF. Further research in the field of kinetics, consideration of multicomponent media, and experimental validation of the model will enable the transition to the design of biomaterials with tailored properties.

1. Zaaba N. F., Jaafar M. A review on degradation mechanisms of poly(lactic acid): Hydrolytic, photodegradative, microbial, and enzymatic degradation // *Polymer Engineering & Science*. 2020. Vol. 60. No. 9. P. 2061-2075 <https://doi.org/10.1002/pen.25511>

2. Tõnsuaadu K. et al. A new perspective on fluorapatite dissolution in hydrochloric acid: Thermodynamic calculations and experimental study // *Inorganics*. 2021. Vol. 9. No. 8. P. 65. <https://doi.org/10.3390/inorganics9080065>

The research was carried out within the state assignment of ministry of science and higher education of the Russian Federation, theme № 075-03-2025-469.